

Models of Disability

05332/05632/17332/17732 Fall 2026

Introduction to Accessibility and Assistive Technology

<https://cmu-intro-to-accessibility.github.io>

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Administrivia

- HW1 was due at 11am today.
- HW2 is due Friday, Sept 4 11:59pm.
 - Yes, it's only 4 days away.
 - But no need to work over the Labor Day holiday!

Models of Disability

- Perspectives are based on a *cognitive authority*
 - *Who* decides what disability means?
 - *Who* decides who is disabled?
- The model of disability to which you subscribe matters!
 - As a designer of Assistive Technology
 - For your users
 - On our institutions and laws

Medical Model of Disability

- Based on science and medicine
- Disability is understood as an individual and/or medical phenomenon that results in limited functioning that is seen as deficient.

Medical Perspective

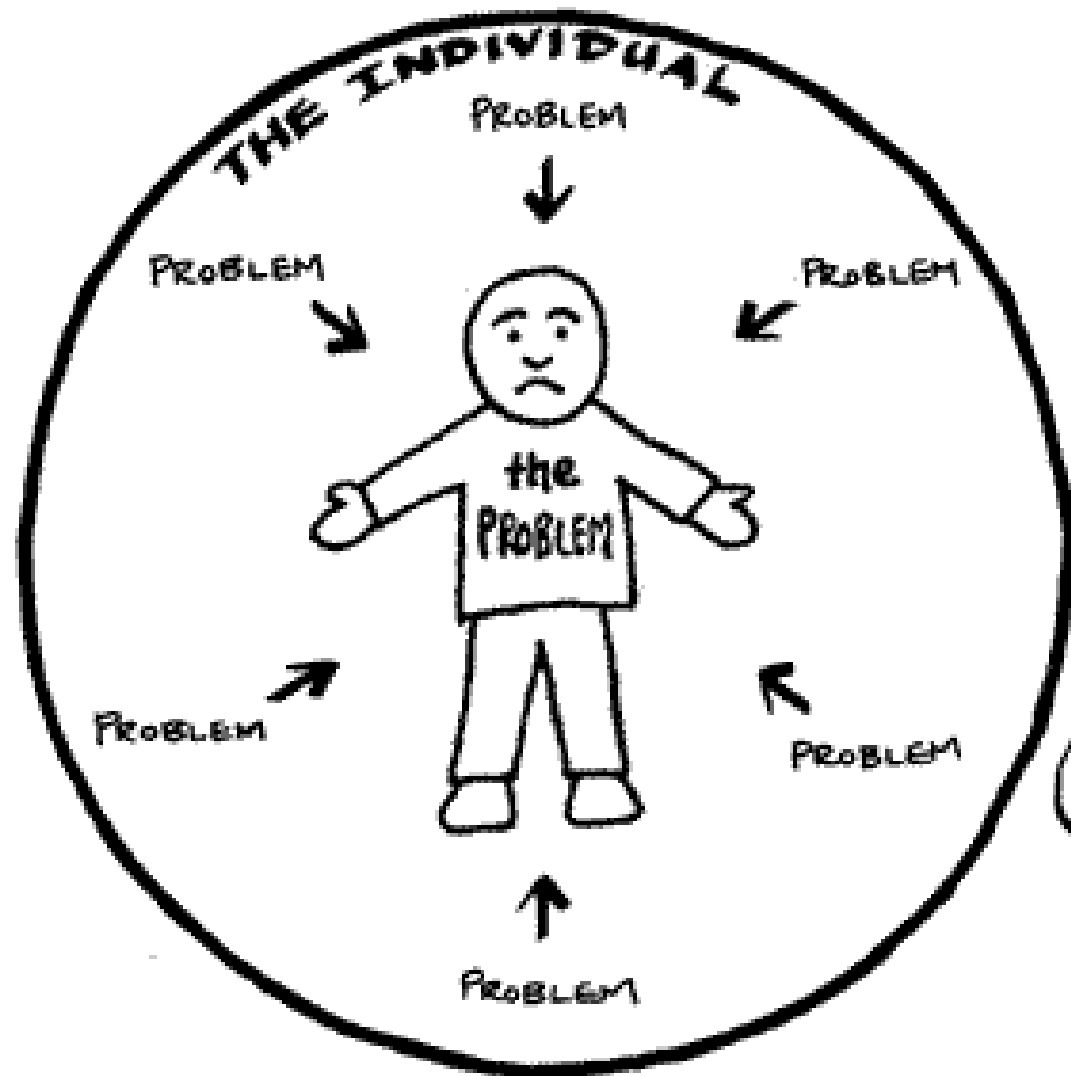
- Closely ties perceptions and definition of disability with “illness”
- Disability is something to be cured or fixed.
 - This will help the individual function in society.
- Emphasizes the value of *normal* functionality.

Aside: The History of *Normal*

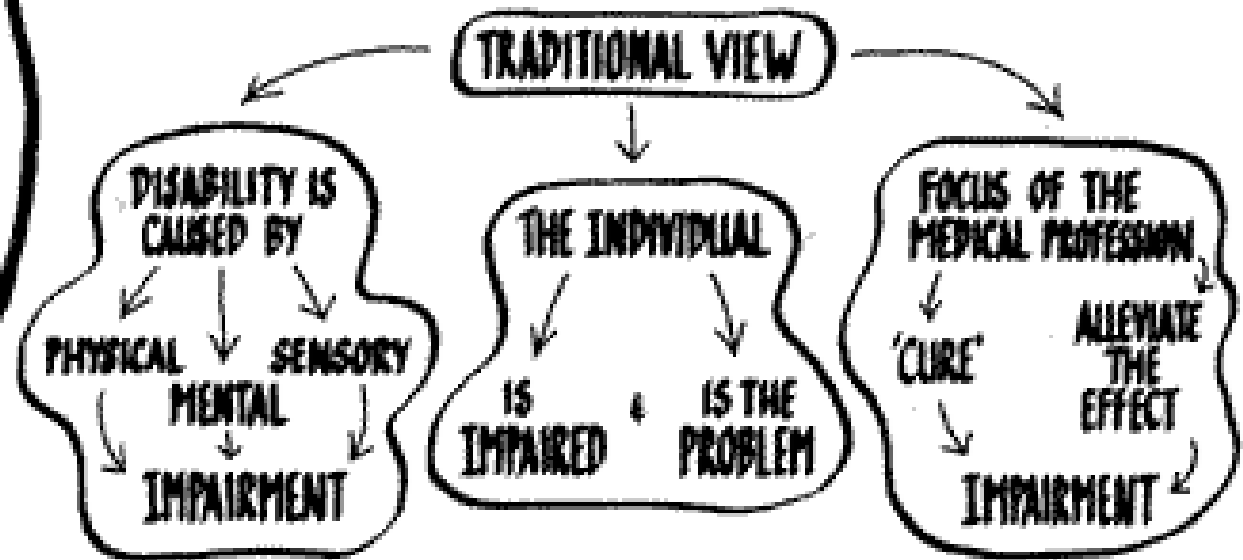
- In early 1900s, insurance companies only wanted to enroll “healthy” people.
 - Pneumonia and tuberculosis more likely to kill underweight people.
- In 1942, Metropolitan Life Insurance Company created standard tables for “ideal” height and weight based on their customers and their health outcomes.
 - These heights and weights fit a normal distribution.
- Average height and weight were considered “ideal,” or *normal*.
- People who were under or overweight *deviated* significantly from the *average*. They became *abnormal*.

From Dan Bouk. [How Our Days Became Numbered: Risk and the Rise of the Statistical Individual](#). University of Chicago Press. 2015.

THE MEDICAL MODEL OF DISABILITY



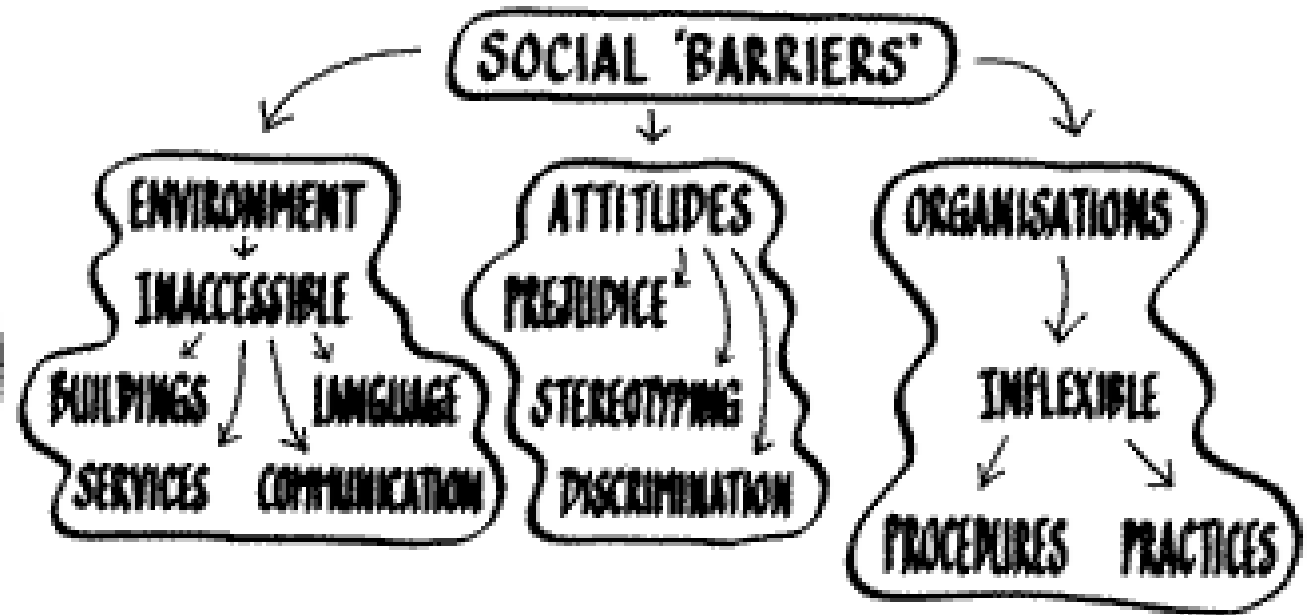
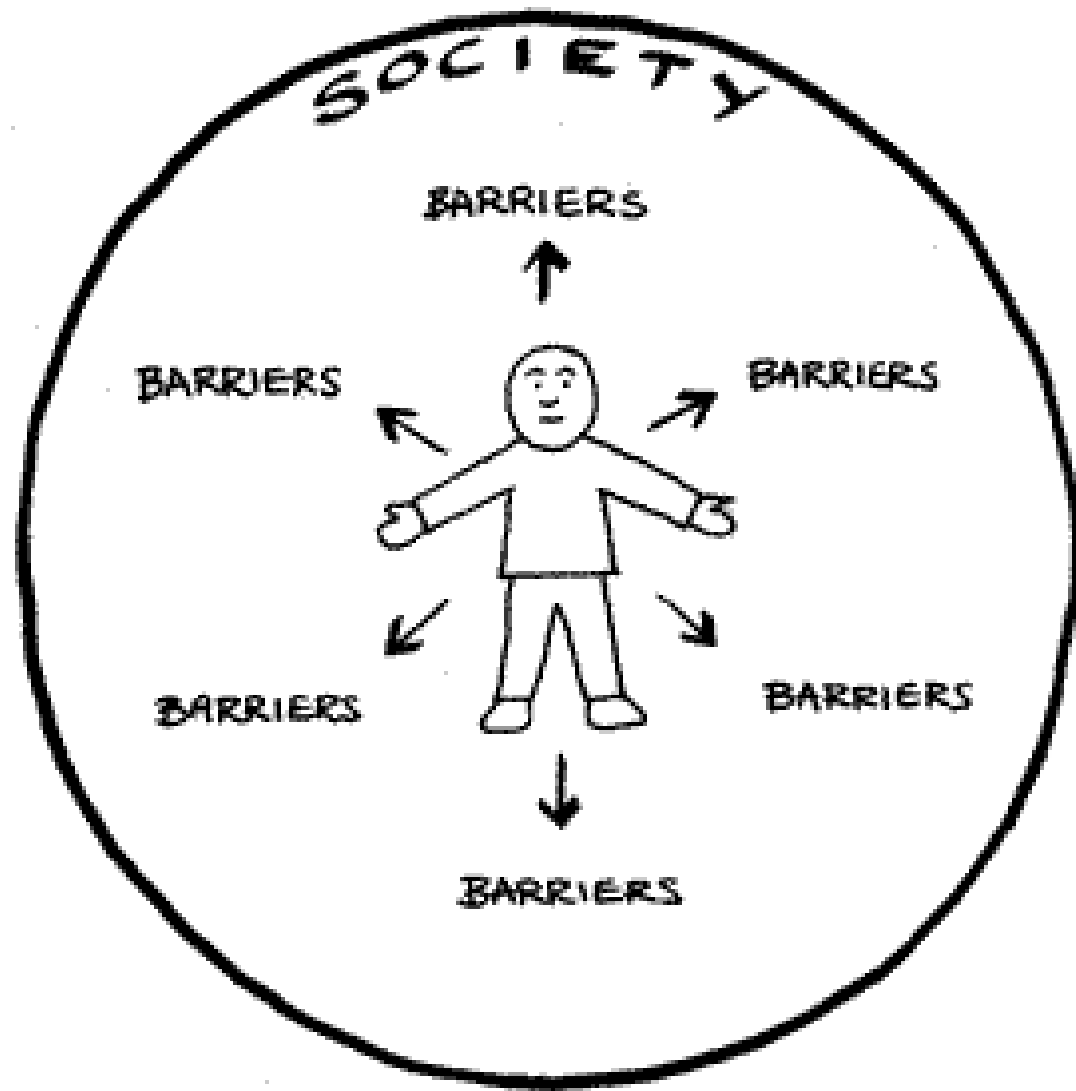
IMPAIRMENTS AND CHRONIC ILLNESS
OFTEN POSE REAL DIFFICULTIES BUT
- THEY ARE NOT THE MAIN PROBLEMS



Social Model of Disability

- Individuals with functional impairments are labeled by others as *disabled*.
 - Society perceives impairments as “a physical or biological condition of a person” meaning that a limb, organ, or function of the body is somehow *negatively* different.
- Thus, disability is a form of *exclusion* propagated by a society that marginalizes or discriminates against people who are seen as being impaired. (Mankoff et al. 2010)

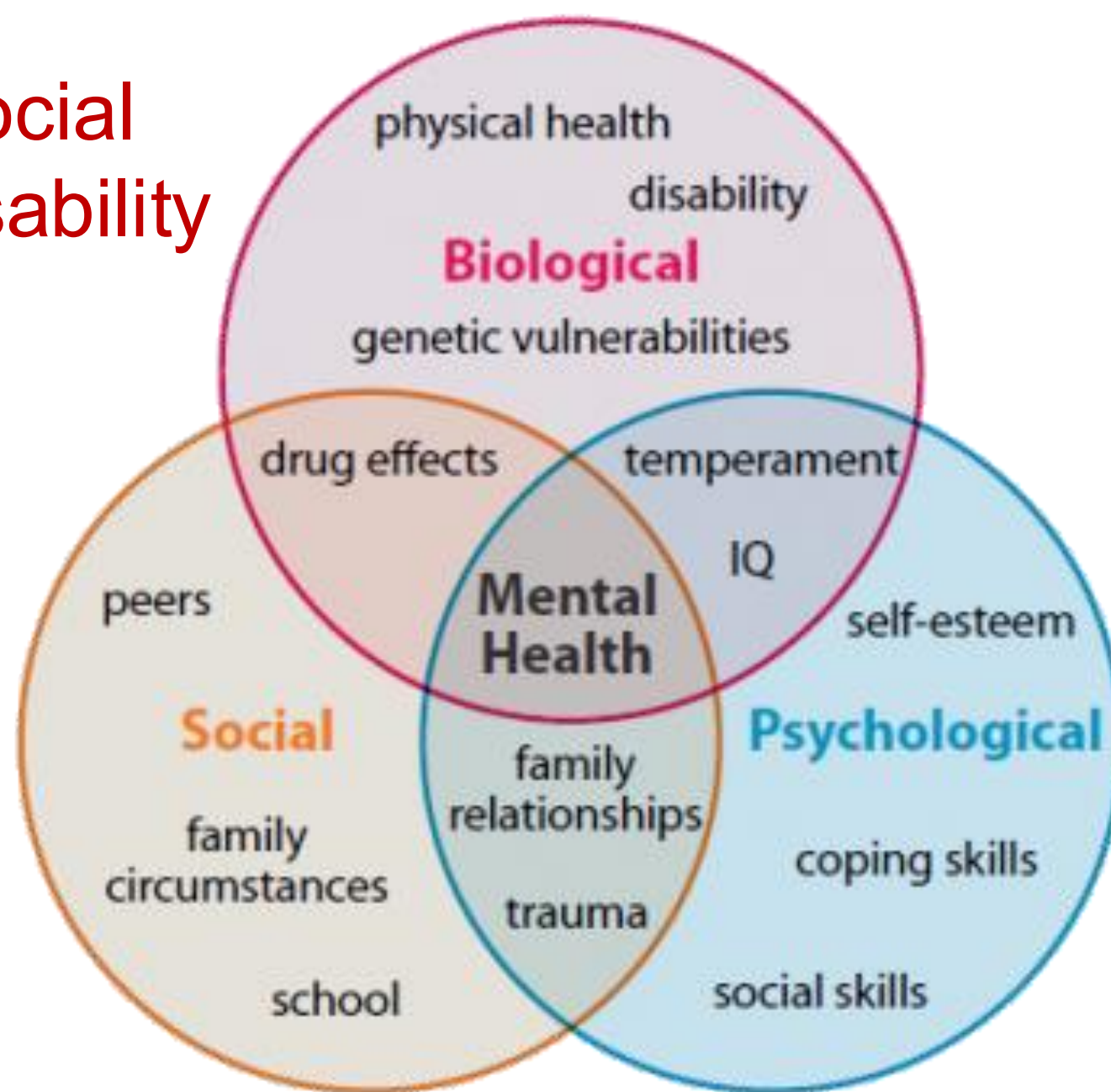
THE SOCIAL MODEL OF DISABILITY



Biopsychosocial Model of Disability

- Understanding a person's condition requires recognizing their physiology, their thoughts, emotions, and behaviors, and their socio-economic and cultural circumstances.
- Commonly used for describing *chronic pain*.

Biopsychosocial Model of Disability



What does this mean for us?

Which model is best?

Medical Model Designs

- Focus on the physical and functional limitations a person demonstrates.
- Designs intend to “fix” impairments
 - Walk down stairs
 - Move around the house
 - Increase typing speed
 - Read the computer screen
 - Help a person communicate



Medical model is useful for scoping

- Especially for programmers and engineers
- There is a functional limitation. Your goal is to enable the function.

Social Model Designs

- Focus on fixing problems of access, oppression, and activism.
- Shift from “cure” to “care.”
- Self-advocacy and peer support lead to full participation in society.
- People with disabilities become the leaders in managing their conditions.
- Solutions include promoting societal change.



Biopsychosocial Model Designs

- Address the whole person *and* environment interaction.
- Remove environmental barriers while also recognizing that a disabled person experiences fatigue, anxiety, stress, and cognitive load.
- This applies to engaging with a world not designed for them as well as engaging with one's own assistive technologies!

All of these models have limitations

Ableism: None consider the potential *positive* aspects of the disabled experience. They often destroy them as part of a solution.

Oversimplification: Disability diagnosis, symptoms, and impairments are not independent nor are they meaningful without context.

Connection: People with disabilities do not exist by themselves. They are interdependent on a large number of support systems, including families, caregivers, doctors, service workers, and friends.

Back to Learning about Disabilities

Hearing Disabilities

- Labels: Deaf, deaf, hard of hearing
- Definitions:
 - deaf: auditory sense channel is unavailable
 - hard of hearing: auditory sensing is limited (threshold > 20 dB)
 - ASL: American Sign Language – a visual language most used by people who are born deaf.
 - Deaf: people who are immersed in the (sign) language and culture of deafness
- Prevalence:
 - 0.25% of USA children born with hearing loss
 - 5% of all people on Earth have hearing loss (WHO)
 - 15% of USA adults over 18 have difficulty hearing.
- Hearing loss increases with age

Kinds of Hearing Loss

- Conductive: Sound prevented from passing through the outer ear or middle ear.
- Sensorineural: Damage to the inner ear
- Mixed: Both conductive and sensorineural
- Usually qualified as loss × pitch (frequency)
 - Mild
 - Severe
 - Profound

Cognition

- Labels: Intellectual developmental disorder (IDD), autism, ADHD, dyslexia, dyscalculia, cerebral palsy, Tourette's, low IQ, dementia, learning disability
- There are many different kinds of disabilities here.

Prevalence of Cognition Disabilities in the USA

- IDD: 8.5%
- Autism: 3%
- ADHD: 9.4%
- Dyslexia: 15%
- Dyscalculia: 3-6%
- Dementia: 0.7%
- Tourette's: 0.6%
- Cerebral Palsy: 0.15%

Kinds of Cognitive Impairments

- Practical Skills:
 - Everyday life skills, Personal care, Self-care
- Conceptual Skills:
 - Memory, Language, Numbers/Math, Decision-Making
- Social Skills:
 - Communication, Social rules

Speech

- Labels: Mute, Dysarthric speech, speech impediment, aphasia, stuttering, cleft palate, laryngitis
- Prevalence:
 - 5% of children have a speech impairment
 - 5% of children stutter for 6 months or more. 1% stutter long-term.

Kinds of Speech Impediments

- Disfluency: Altered flow of a person's speech
- Orofacial myofunctional: Using or moving your face, mouth, tongue muscles
- Speech sound: Ability to produce clear sounds
- Voice: Quality of your vocal sounds

Mobility or Motor

- Labels: Mobility impairment, motor impairment, wheelchair user, prosthetic user, amputee,
- Definitions: Motor skills involve motor, sensory, and cognitive abilities.
 - **Fine Motor Control (Dexterity)** - small muscle movements often the hands and fingers.
 - **Gross Motor Control (Mobility)** - typically refers to the larger muscle groups including the legs and impact walking and more generalized movement of the body
 - **Range of Motion** - required for different tasks may be vastly different
 - **Loss of Limb/Paralysis** – body parts may be missing or brain cannot control them.

Prevalence of Mobility Disabilities

6.1% of US Adults (21 million) have an ambulatory motor impairment. 12.1 million are over 65. ([US Census Bureau](#))

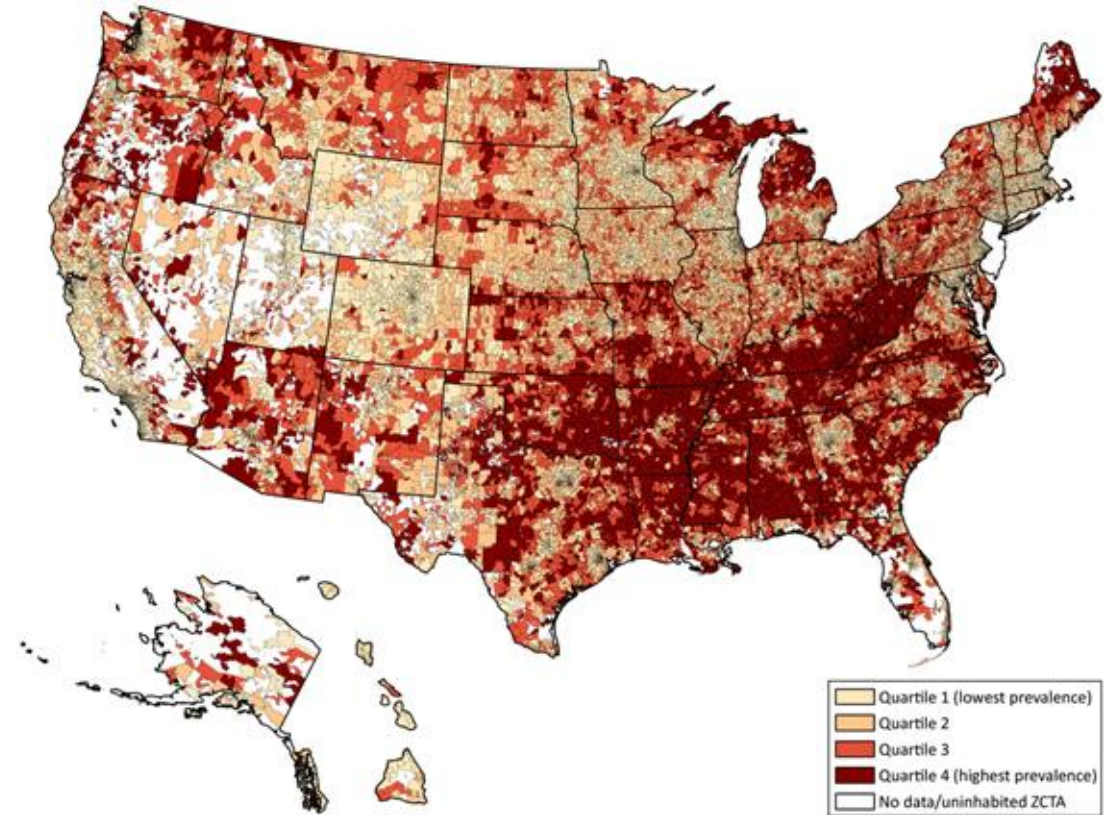
5.5 million people in USA use a wheelchair. ([US DOT, Feb 29, 2024](#))

Chronic Illness

- Labels: Chronically ill, chronic health conditions
- Definition:
 - Range of conditions and diagnoses that impact functioning and are not expected to go away or be immediately fatal.
 - Symptoms alter one's daily life.
- Prevalence:
 - 24.6% of US adults have 1 of 10 specific chronic conditions.
 - 27.2% of US adults have 2 or more of 10 specific chronic conditions.
 - Chronic conditions: hypertension, coronary heart disease, stroke, diabetes, cancer, arthritis, hepatitis, kidney failure, asthma, emphysema.

Prevalence of Chronic Illness

- Geographic distribution of chronic illness is *not* random.
- Chronic illness in the Southeast correlates strongly with decades of social and economics policies (e.g. low minimum wage, non-expansion of Medicaid), concentrated poverty, lack of access to affordable healthcare.



https://www.cdc.gov/pcd/issues/2024/23_0267.htm

Impairments Caused by Chronic Illness

- Physical
 - Pain, fatigue, discomfort, dizziness, nausea
- Mental
 - People are disbelieved (gaslighted!), cautious, hyper-vigilant about controlling symptoms
- Dynamic variability in ability, changing day to day
 - Sometimes after a particular activity, like eating a large meal!
 - Are the consequences worth it?